

P2P Dollars as an Early Warning System: Exchange Market Pressure, Panic Premia, and the Currency Stress Cycle in Bolivia

Working Paper WP/2026/01

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February 2026 (updated daily)

JEL Codes: F31, F37, G15, O16

Abstract

I construct a real-time Exchange Market Pressure (EMP) index for Bolivia using 543 daily observations from Binance peer-to-peer (P2P) cryptocurrency order books, covering August 2024 through February 2026. Bolivia’s boliviano has been officially fixed at BOB 6.96/USD since 2011, but an active parallel dollar market operates openly on Binance P2P, where agents buy and sell Tether (USDT)—a dollar-pegged stablecoin—at shadow prices that deviate

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substantially from the official rate. These deviations, along with the bid-ask spread and order-book depth, contain real-time information about exchange market stress that traditional macroeconomic indicators cannot provide.

The EMP index aggregates three standardised microstructure signals via principal component analysis: (i) the shadow rate gap between the P2P mid-price and the official peg; (ii) the bid-ask spread, which I interpret as a *panic premium* measuring the round-trip cost of informal dollar access; and (iii) liquidity stress from order-book depth. The first principal component explains 55.8% of joint variance and loads primarily on the shadow gap (0.817) and panic premium (0.523).

The 18-month sample reveals a complete stress cycle: a sustained period of moderate pressure through early 2025 (shadow premium near 50%), an acute crisis in April–May 2025 (premium reaching 141%, EMP peaking at $+3.72\sigma$), a gradual correction through mid-2025, and a striking normalisation by early 2026 (premium at 31%, EMP at -1.33σ). The EMP index displays strong persistence ($\text{AR}(1) = 0.947$) and clear regime structure: once EMP enters the high-stress regime it remains there with 78.9% probability.

An expanding-window logistic early-warning model achieves out-of-sample AUC of 0.996 (14-day) and 0.999 (30-day), and liquidity stress alone achieves in-sample AUC of 0.928 for the 30-day horizon. A one-standard-deviation increase in liquidity stress multiplies the 30-day depreciation probability by a factor of 57. The model identifies a key asymmetry: spread and depth are negatively correlated ($r = 0.387$), so markets can be simultaneously expensive (high spread) and liquid (high depth) in crisis episodes, a pattern that a price-only indicator would miss.

A Dynamic Factor Model (DFM) formalises why composite aggregation destroys predictive content. The DFM variance decomposition finds that the three

observables are structurally distinct: the shadow rate gap is 99.7% common factor, while liquidity stress is 99.6% idiosyncratic—orthogonal to the EMP factor but the single best predictor of future depreciation. Combining all three into a composite mixes a coincident price signal with an orthogonal leading liquidity signal, diluting the latter’s predictive power. The DFM’s one operational contribution is a formal confidence band around the EMP estimate (mean $\pm 0.89\sigma$ at 95%), useful for policy communication. Rolling PCA and Markov-switching models add no predictive value over the static baseline.

The practical contribution is a freely accessible, daily-updated monitoring platform combining Supabase, GitHub Actions, and GitHub Pages, demonstrating that complete early-warning infrastructure for a sovereign exchange rate can be built and maintained at near-zero marginal cost from open data sources.

Keywords: exchange market pressure; P2P cryptocurrency; early warning systems; Bolivia; shadow exchange rate; capital controls; informal markets; fixed exchange rate regimes; order-book microstructure.

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1 Introduction

1.1 Motivation

Currency crises are among the most disruptive macroeconomic events in developing economies, capable of wiping out years of reserve accumulation, triggering debt restructurings, and compressing real wages through import-price inflation (Kaminsky et al., 1998). The academic and policy literature on early warning systems (EWS) for currency crises is extensive, yet it faces a fundamental constraint: the macroeconomic indicators that best predict crises—reserve levels, current account balances, real effective exchange rates—arrive at monthly or quarterly frequency, typically with additional publication lags of one to three months (Berg and Patillo, 1999; Frankel and Saravelos, 2012). By the time a standard EWS model produces a warning signal, market participants may have already acted on the information, reserves may have been substantially depleted, and the window for preventive policy action may have closed.

This paper proposes a different approach. In economies with active informal foreign exchange markets, the shadow exchange rate contains real-time information about the demand-supply balance for foreign currency that official statistics cannot provide until months later. The shadow rate is not a noisy proxy for true pressure: it *is* the market’s direct assessment of the value of a dollar at the margin, continuously updated by the decisions of thousands of agents who have direct knowledge of local dollar demand and supply conditions.

1.2 The P2P Revolution in Currency Markets

A development of the past five years has made shadow exchange rate data unusually accessible: the migration of informal dollar trading to peer-to-peer cryptocurrency platforms. In countries with strict capital controls—including Bolivia, Argentina, Venezuela, Cuba, and Zimbabwe—local agents have increasingly turned to platforms like Binance P2P to buy and sell Tether (USDT), a stablecoin pegged one-to-one to the US dollar. These transactions occur at freely negotiated prices, are publicly observable in the platform’s order book, and are recorded at near-continuous frequency.

The Binance P2P order book provides three distinct information signals:

1. **The shadow price itself:** how many bolivianos agents are willing to pay (or accept) for one USDT above the official rate.
2. **The bid-ask spread:** the gap between ask and bid prices measures market friction, search costs, and counterparty risk—all of which increase when devaluation pressure intensifies.
3. **Order-book depth:** the total volume of USDT available for immediate purchase or sale reveals market liquidity—falling depth signals that market makers are withdrawing, often a precursor to price spikes.

Together, these three signals provide a richer picture of market conditions than any single price indicator could offer.

1.3 Bolivia as a Laboratory

Bolivia provides an unusually clean setting for this analysis for four reasons. First, the official peg at BOB 6.96/USD has been *immovable* since 2011: there have been no official adjustments in the entire sample, which means that any deviation of the P2P

price from 6.96 is entirely attributable to informal market forces rather than to regime changes. Second, the P2P market has been active and relatively liquid throughout the sample, providing sufficient observations to estimate models. Third, the Bolivian case involves a genuine and escalating reserve constraint: BCB gross international reserves fell from USD 15 billion in 2014 to approximately USD 1.7 billion by 2024 (IMF, 2024), providing fundamental pressures that the P2P market should be reflecting. Fourth, Bolivia’s P2P market experienced a dramatic stress episode in 2025—the shadow premium reaching 141%—which provides the kind of sharp variation needed to identify early-warning relationships.

1.4 Preview of Main Results

The data reveal a complete and well-structured stress cycle between August 2024 and February 2026. The shadow premium averaged 50% during the pre-escalation phase (August–December 2024), accelerated to 106% by April–May 2025, peaked at 141% on 15 May 2025, and has since corrected to 31% as of February 2026. Over this full cycle:

- The EMP index peaked at $+3.72\sigma$ in May 2025 and currently stands at -1.33σ , its lowest level in the sample.
- The bid-ask spread (panic premium) ranged from 6.1% to 33.2%, providing strong regime-identifying variation.
- Order-book depth nearly doubled from the crisis trough to the current normalisation period.
- Liquidity stress is a remarkably powerful predictor of medium-term shadow depreciation: the in-sample AUC of z_ℓ for the 30-day horizon is 0.928, and out-of-sample expanding-window AUC is 0.999.

The composite EMP index, while useful as a summary measure, underperforms

individual components as a predictive variable—a result that argues for monitoring all three signals in parallel rather than compressing them into a single score.

1.5 Structure of the Paper

Section 2 provides institutional background on Bolivia’s exchange rate regime and the P2P market. Section 3 describes the data in detail. Section 4 discusses the microstructure theory connecting P2P order books to exchange market pressure. Section 5 constructs the EMP index step by step. Section 6 presents the early warning model. Section 7 narrates the four phases of the stress cycle visible in the data. Section 8 reports the full empirical results with all tables. Section 9 discusses robustness. Section 10 extends the analysis to state-space alternatives—a Dynamic Factor Model, rolling PCA, and a Markov-switching AR(1)—and uses the DFM variance decomposition to provide a structural justification for monitoring the three components separately rather than as a single composite. Section 11 addresses limitations. Section 12 concludes. Appendices cover the data pipeline, variable definitions, and figures.

2 Institutional Context

2.1 Bolivia’s Fixed Exchange Rate and Reserve Constraint

Bolivia introduced exchange rate stability as a cornerstone of its post-hyperinflation macroeconomic framework in the late 1980s. After a decade of crawling-peg adjustments, the rate was fixed at BOB 6.96/USD in November 2011, a level that has remained unchanged for over 14 years.

The peg was initially backed by a large reserve cushion. During the commodity

boom, Bolivia’s gas exports generated current account surpluses reaching 12% of GDP, BCB reserves accumulated to USD 15 billion (roughly 18 months of imports), and the government ran fiscal surpluses. The combination of high reserves, low inflation, and exchange rate stability established Bolivia as an unusually well-managed developing economy (IMF, 2024).

The commodity price reversal after 2014 exposed the vulnerability of this model. Bolivia’s gas exports declined in both volume and price. The government maintained public investment and social spending by drawing down reserves. Persistent fiscal deficits averaged 7–8% of GDP between 2016 and 2023. By 2024, BCB gross international reserves had fallen to approximately USD 1.7 billion—less than two months of imports and far below IMF adequacy benchmarks for a fixed-rate economy.

Critically, the BCB has not officially devalued the boliviano despite this reserve erosion. Instead, it has rationed dollar access: official exchange transactions are subject to documentation requirements, commercial bank dollar purchases are limited, and there have been reports of delays and restrictions on dollar access in the banking system. This rationing creates the conditions for a parallel market premium: when official dollars are scarce and restricted, agents who need dollars turn to informal channels, and the price they are willing to pay in those channels reflects both the fundamental demand-supply imbalance and the convenience premium of unrestricted access.

2.2 The Bolivian P2P Dollar Market

Informal dollar trading in Bolivia has a long history, traditionally conducted through street-level cambistas (money changers) and informal broker networks. Starting around 2020–2021, a significant share of this activity migrated to Binance’s peer-to-

peer trading platform. Binance P2P is a matching service that connects buyers and sellers without Binance itself taking a position: sellers post advertisements specifying their price and available quantity, buyers browse listings and initiate transactions, and Binance holds the USDT in escrow until the fiat transfer is confirmed.

The platform is formally accessible in Bolivia and has operated without direct regulatory prohibition. This accessibility—combined with the global liquidity of the USDT market—has attracted a large number of Bolivian participants. On any given day, there are hundreds to thousands of active USDT/BOB advertisements, creating a reasonably liquid informal market at granular price frequencies.

Two prices, one economy. Bolivia now operates with two effectively distinct dollar prices:

1. **Official rate (dólar oficial):** BOB 6.96/USD, available through the BCB and licensed commercial banks, subject to rationing and documentation requirements.
2. **P2P shadow rate:** the market-clearing price for USDT/BOB on Binance P2P, freely determined, continuously observable, and reflecting genuine marginal willingness-to-pay for unrestricted dollar access.

The gap between these two prices—which I call the shadow premium—is the central object of study in this paper. As of February 2026, the shadow premium stands at 31%, meaning that agents are willing to pay BOB 9.15 for a dollar they could in principle obtain at BOB 6.96 through official channels. This persistent premium reflects the combination of rationing (not everyone can access the official rate), transaction costs and delays in official channels, and a pure risk premium compensating sellers for the informal nature of the transaction.

3 Data

3.1 Sources

The dataset is assembled from two complementary sources that together cover August 2024 through February 2026 at approximately 30-minute intraday frequency.

Source 1: Historical P2P price series (Aug 2024 – Nov 2025). An intraday series of USDT/BOB bid and ask prices at approximately 30-minute intervals. Each observation provides: timestamp, quantity-weighted average price (VWAP) for each side, price range statistics (low, high, median), total tradeable quantity, number of active advertisements, and the Herfindahl-Hirschman Index (HHI) of advertiser concentration. This source covers 5 August 2024 through 27 November 2025.

Source 2: Binance P2P order-book snapshots (Nov 2025 – present). Full advertisement-level data from the public Kaggle dataset `andreschirinos/p2p-bob-exchange`, downloaded daily via the `kagglehub` Python API. Each snapshot contains, for every active USDT/BOB advertisement: advertiser ID, quoted price (BOB/USDT), available quantity (USDT), and timestamp. VWAP and depth are computed per equation (1) in Section 3.3. This source updates continuously.

3.2 Sample Construction and Merge

The two sources overlap from 28 November to 7 December 2025. A direct comparison of the VWAP series in this window confirms near-perfect agreement: the correlation between the ask-side VWAPs from the two sources exceeds 0.999, validating the merge. The merged dataset uses Source 1 for all observations prior to 28 November 2025 and

Source 2 thereafter.

The final merged sample covers **29 August 2024 to 22 February 2026**, comprising:

- Approximately 22,000 intraday (30-minute) observations
- **543 daily aggregates** after end-of-day aggregation

3.3 Variable Construction

Quantity-weighted average price (VWAP). For each snapshot, the VWAP on each side is:

$$\hat{p}_t^s = \frac{\sum_i q_{it}^s p_{it}^s}{\sum_i q_{it}^s}, \quad s \in \{\text{ask}, \text{bid}\}, \quad (1)$$

where p_{it}^s is the quoted price and q_{it}^s is the available quantity for advertiser i on side s . The VWAP is the appropriate price concept for assessing what a representative agent—one who wants to execute a quantity-weighted average trade, not just transact at the best price—would pay or receive.

Sign convention. The ask price ($s = \text{ask}$) is quoted by agents offering to *sell* USDT (i.e. they are willing to receive bolivianos and hand over USDT). The bid price ($s = \text{bid}$) is quoted by agents offering to *buy* USDT (they offer bolivianos in exchange for USDT). Since agents buying dollars through the platform pay the ask price and receive the bid price, we have $p_t^{\text{ask}} > p_t^{\text{bid}}$ by arbitrage, and the ask is the price relevant for measuring the cost of dollar access.

Shadow mid-price and premium.

$$p_t^{\text{mid}} = \frac{1}{2}(p_t^{\text{ask}} + p_t^{\text{bid}}), \quad (2)$$

$$\text{prem}_t = \frac{p_t^{\text{mid}} - 6.96}{6.96} \times 100\%. \quad (3)$$

Bid-ask spread (panic premium).

$$\tau_t = \frac{p_t^{\text{ask}} - p_t^{\text{bid}}}{p_t^{\text{mid}}} > 0. \quad (4)$$

τ_t measures the round-trip cost of informal dollar access: an agent who buys USDT at the ask and immediately resells at the bid loses $\tau_t \times 100\%$ of the transaction value.

Log order-book depth.

$$d_t = \ln(\min(D_t^{\text{ask}}, D_t^{\text{bid}})), \quad D_t^s = \sum_i q_{it}^s. \quad (5)$$

Taking the minimum of the two sides captures the binding constraint on immediate transaction size. A low value of d_t means that a large order would quickly exhaust the book on at least one side.

3.4 Descriptive Statistics

Table 1 presents full summary statistics for the key variables over the 543-day sample. Several features stand out.

Right-skewed shadow premium. The distribution of the mid-price is positively skewed (skewness 0.87) with excess kurtosis of -0.15 . The Jarque-Bera statistic of 68.9 decisively rejects normality. The 95th percentile (BOB 15.53) is nearly twice the 5th percentile (BOB 9.28), reflecting the asymmetry between the acute crisis peak and the recent normalisation trough. The average premium over the full sample is 67.7%, but this masks enormous variation: monthly averages range from 42% (November 2025) to 141% (May 2025).

Fat-tailed spread distribution. The bid-ask spread also departs strongly from normality (JB = 22.9). The median spread of 13.9% is substantially below the 95th percentile of 23.0%, consistent with occasional sharp widenings during stress episodes. The minimum spread of 6.1% (observed in February 2026) represents the closest the market has come to “normal” functioning during the sample.

Depth as a regime indicator. Log depth ranges from 11.56 to 13.87—a factor of $e^{2.31} \approx 10$ -fold in levels. The most notable feature is that depth was *not* at its minimum during the May 2025 price peak. Rather, depth reached its minimum during the pre-escalation phase, when the market was thin and dominated by a few large advertisers. As the premium rose and the market attracted more participants, depth increased—before collapsing again as the crisis intensified. This non-monotonic relationship between price level and depth is a key finding that motivates the multi-signal approach.

Table 1: Summary Statistics, Full Sample (543 daily observations)

	Mean	Std	Skew	Min	p_5	p_{50}	p_{95}	Max
Mid price (BOB/USDT)	11.67	1.86	0.87	9.02	9.28	10.89	15.53	16.79
Shadow gap (BOB)	4.71	1.86	0.87	2.06	2.32	3.93	8.57	9.83
Shadow premium (%)	67.7	26.7	0.87	29.6	33.3	56.5	123.3	141.2
Spread τ (%)	14.78	4.98	0.34	6.13	7.62	13.94	23.03	33.15
Log depth	12.61	0.57	0.11	11.56	11.75	12.63	13.57	13.87
EMP index (σ)	−0.06	1.16	0.53	−1.84	−1.53	−0.43	1.90	3.72
N	543 (29 August 2024 – 22 February 2026)							

Notes. Daily aggregates from ≈ 30 -minute intraday snapshots. Mid price = $(p^{\text{ask}} + p^{\text{bid}})/2$. Shadow gap = mid − 6.96. Shadow premium = gap/6.96 $\times$ 100. Spread = $(p^{\text{ask}} - p^{\text{bid}})/p^{\text{mid}}$. Log depth = $\ln(\min(D^{\text{ask}}, D^{\text{bid}}))$. EMP is the first principal component of standardised (z -scored) gap, spread, and log depth. Skewness computed using the standard sample skewness formula. JB (Jarque-Bera) statistics reject normality for all variables at $p < 0.001$.

4 Market Microstructure and Exchange Pressure

4.1 Why P2P Order Books Reflect Currency Stress

The connection between P2P order-book microstructure and currency stress follows from three basic principles of market microstructure.

1. The shadow price as a sufficient statistic for marginal demand. In any market with a binding price ceiling—here, the official rate acts as a notional ceiling, while rationing creates effective scarcity above that level—the shadow price reveals the marginal agent’s willingness to pay. When the BCB rations official dollar access, agents who cannot satisfy their dollar demand through official channels turn to the P2P market. The P2P price is therefore the marginal cost of a dollar for the marginal agent, and its deviation from 6.96 measures the magnitude of excess demand in equilibrium.

2. Bid-ask spread as risk premium and search cost. In a market-maker model, the bid-ask spread compensates market makers for three costs ([Glosten and Harris, 1988](#)): order processing costs, inventory costs, and adverse selection costs (the risk that the counterparty has superior information). In the Bolivian P2P context, the dominant components are likely inventory costs (the risk of holding a devaluing asset) and adverse selection (the risk that buyers are purchasing dollars precisely because they anticipate imminent devaluation). Both components increase when devaluation pressure is high: market makers widen their spread to protect against being caught on the wrong side of a price move. Thus the spread τ_t measures market makers’ collective assessment of devaluation risk, not just transaction costs.

3. Order-book depth as a predictor of future volatility. A large literature in market microstructure documents that falling order-book depth predicts future price volatility and adverse price movements (Glosten and Milgrom, 1985). The economic intuition is that a thin book is fragile: a relatively small market order can move prices substantially. When informed agents believe a devaluation is likely, they withdraw sell-side liquidity (they do not want to sell USDT cheaply today if the boliviano will be worth less tomorrow), creating a thin ask book that is then easily pushed to higher prices by arriving buyers. This is the “inventory withdrawal” mechanism: depth falls *before* prices spike, providing a forward-looking signal that price-level indicators miss.

4.2 The Panic Premium: A Closer Look

The bid-ask spread τ_t deserves special attention as an indicator. Consider a Bolivian firm that needs USD 100,000 to pay a foreign supplier. It can attempt to obtain dollars at the official rate (BOB 696,000 at 6.96) or through the P2P market. At the mid-point of the P2P market (BOB 11.67 on average), the cost is approximately BOB 1,167,000—a 67.7% premium. But the ask price is *above* the mid-point by half the spread. If the spread is 15%, the ask is:

$$p_t^{\text{ask}} = p_t^{\text{mid}} \times (1 + \tau_t/2) \approx 11.67 \times 1.075 = 12.55 \text{ BOB/USDT}.$$

So the total all-in premium paid by the firm is not 67.7% but rather $12.55/6.96 - 1 = 80.3\%$. The spread adds approximately 7.5 percentage points to the cost of dollar access beyond the mid-price premium. During the May 2025 crisis, with a mid-price of BOB 16.79 and a spread of 33%, the all-in premium reached: $16.79 \times 1.165/6.96 - 1 \approx 154\%$.

This “all-in premium” calculation illustrates why I call τ_t the *panic premium*: it

measures not just market illiquidity but the additional cost imposed on agents who *must* transact immediately and cannot wait for better prices. During genuine panic episodes, those who most urgently need dollars—to pay foreign obligations, to hedge savings against inflation, to emigrate—face the widest spreads.

4.3 Asymmetric Information and the Depth-Spread Correlation

The correlation between spread and depth in the sample is +0.387 (Table 10). This positive correlation—wide spreads *tend* to be associated with deeper books—is at first glance counterintuitive. Standard microstructure theory predicts that illiquid markets (low depth) have wide spreads.

The resolution lies in the crisis dynamics. During the acute May 2025 crisis, the shadow premium surged and both the spread and the total number of advertisers increased simultaneously: the high premium attracted *more* sellers of USDT into the market (expanding depth) at the same time as buyers panicked (widening spreads). The positive spread-depth correlation therefore reflects a supply response to price signals: high premiums induce entry of dollar sellers, thickening the book even as spreads remain wide from buyer-side pressure.

This asymmetry has a direct implication for EWS design: a model that monitors only the price level (gap) or only the spread would miss the within-crisis signal embedded in the joint dynamics of spread and depth. The PCA approach used below captures this joint variation optimally.

5 The Exchange Market Pressure Index

5.1 The Original EMP Framework and Its Limitations

The EMP concept originates with [Girton and Roper \(1977\)](#), who defined exchange market pressure as the total pressure on a currency as a weighted sum of the exchange rate change and the official reserve change. The intuition is that pressure manifests through one or both channels: if the central bank defends the rate, pressure shows up as reserve losses; if it allows adjustment, pressure shows up as depreciation. [Eichengreen et al. \(1996\)](#) extended this to a three-way decomposition (exchange rate, reserves, interest rate differential).

The standard EMP framework faces two problems in the Bolivian context. First, reserve data are published monthly with a further lag, eliminating any real-time monitoring capacity. Second, Bolivia has no meaningful policy interest rate that responds to exchange pressure. Third, the official exchange rate has not changed since 2011, so the rate-change component is identically zero.

The P2P microstructure approach resolves all three problems: the shadow price, spread, and depth are all observable at 30-minute frequency with essentially zero lag.

5.2 Component Definitions and Orientation

Definition 1. *The three EMP components are:*

$$g_t = p_t^{mid} - 6.96 \quad (\text{shadow rate gap}), \quad (6)$$

$$\tau_t = \frac{p_t^{ask} - p_t^{bid}}{p_t^{mid}} \quad (\text{panic premium}), \quad (7)$$

$$\ell_t = -\ln(\min(D_t^{ask}, D_t^{bid})) \quad (\text{liquidity stress}), \quad (8)$$

where $D_t^s = \sum_i q_{it}^s$ is total advertised depth on side s .

All three components are oriented so that higher values indicate greater pressure: g_t rises with the shadow premium; τ_t rises when the market is more expensive to enter; and ℓ_t rises when the market is thinner (note the negative sign, so $\ell_t = -\ln(\text{depth})$ is high when depth is low). This orientation convention ensures that all three components are positively associated with stress, which is important for the PCA interpretation.

5.3 Standardisation

Each raw component is standardised to zero mean and unit variance using full-sample statistics:

$$z_{kt} = \frac{c_{kt} - \bar{c}_k}{\hat{\sigma}_{c_k}}, \quad k \in \{g, \tau, \ell\}, \quad (9)$$

where $\bar{c}_k$ and $\hat{\sigma}_{c_k}$ are the sample mean and standard deviation. The standardised series z_{kt} has an important substantive interpretation: it measures the current reading of each component in units of “standard deviations above the sample mean.” A value of $z_{g,t} = 2$ means the shadow gap is two standard deviations above its historical average—a genuinely unusual reading.

Because the standardisation uses *full-sample* statistics, it is implicitly backward-looking: the interpretation of each z -score depends on the history observed in the sample. As the sample extends, the full-sample statistics will evolve, and the standardised values will be recalibrated accordingly. The live pipeline handles this automatically through daily re-estimation.

5.4 Principal Component Analysis

Definition 2. *The EMP index is the first principal component of the vector $\mathbf{z}_t = (z_{g,t}, z_{\tau,t}, z_{\ell,t})^\top$:*

$$EMP_t = \mathbf{w}_1^\top \mathbf{z}_t, \quad (10)$$

where $\mathbf{w}_1$ is the eigenvector corresponding to the largest eigenvalue of the sample covariance matrix of $\mathbf{z}_t$, with the normalisation that the loading on $z_{g,t}$ is positive.

Why PCA? PCA extracts the single linear combination of the three signals that explains the maximum fraction of their joint variance. This is optimal under the following assumption: if all three components respond to a single underlying factor (“exchange market pressure”), the first PC is the minimum-variance estimator of that factor. The EMP interpretation requires that the three components be positively correlated in the data—which they are ($r_{g\tau} = 0.663$, $r_{g\ell} = 0.051$, $r_{\tau\ell} = 0.387$ in Table 10)—so the first PC captures predominantly the common stress component.

Alternative aggregation. An alternative to PCA is equal-weight averaging: $EMP_t^{\text{ew}} = (z_{g,t} + z_{\tau,t} + z_{\ell,t})/3$. In Section 9, I show that the PCA and equal-weight versions are highly correlated ($r = 0.96$) and deliver nearly identical regime classifications. The advantage of PCA is that it assigns weights proportional to each signal’s predictive content rather than imposing equal weights by assumption.

5.5 PCA Results

Table 2 reports the full PCA decomposition. The first principal component (EMP) explains 55.8% of joint variance, with loadings of +0.817 (shadow gap), +0.523 (panic premium), and −0.242 (liquidity stress).

Table 2: Principal Component Analysis: Eigenvectors and Variance Shares

	Loadings			Variance explained	Cumulative
	z_g (gap)	z_τ (spread)	z_ℓ (liq. stress)	(%)	(%)
PC1 (EMP)	+0.817	+0.523	−0.242	55.8	55.8
PC2	+0.346	−0.110	+0.932	32.2	88.0
PC3	−0.461	+0.845	+0.271	12.0	100.0

Notes. PCA estimated on 543 daily observations (29 Aug 2024 – 22 Feb 2026). All three components standardised before PCA. PC1 is oriented so that the loading on z_g is positive. The negative loading on z_ℓ in PC1 reflects a positive sample correlation between depth and price level: depth *increased* as the premium rose during April–May 2025, as higher prices attracted new seller-side entrants. Deeper depth is therefore associated with *higher* prices in this sample, implying that the “liquidity stress” component (negative depth) loads negatively on the price-spread dimension captured by PC1. PC2 is almost entirely a depth factor (loading 0.932), orthogonal to price-spread dynamics.

The negative loading on z_ℓ warrants careful interpretation. The signed component in equation (8) is *liquidity stress*: a positive $z_{\ell,t}$ means depth is *below* average, which in isolation should signal more pressure. The negative PC1 loading on this component therefore means that *below-average depth is associated with lower EMP*, which sounds wrong. The resolution is the supply response described in Section 4: the full-sample correlation between depth and price is positive ($r = 0.051$), and the correlation between depth and spread is also positive ($r = 0.387$). So when prices are high (high stress), depth tends to be higher, not lower—and the PCA correctly reflects this joint distribution by assigning a negative weight to z_ℓ in the price-spread direction.

PC2, which explains 32.2% of variance, loads almost entirely on depth (+0.932), capturing variation in order-book liquidity that is orthogonal to price and spread. This component is economically distinct: it identifies days when the market is unusually deep or shallow relative to what price and spread conditions would predict, which may capture advertiser concentration dynamics or seasonal patterns.

5.6 EMP Dynamics and Persistence

The EMP index displays strong positive serial correlation. Table 3 reports autocorrelations at selected lags.

Table 3: Autocorrelation Function of the EMP Index

Lag (days)	1	2	3	5	7	10	14	30
ACF	0.947	0.923	0.905	0.882	0.866	0.844	0.833	0.717

Notes. Sample autocorrelations of the daily EMP index. All values significant at $p < 0.001$ (Ljung-Box test). The slow decay is consistent with a near-unit-root process: EMP regimes are highly persistent once established.

The first-order autocorrelation of 0.947 is high, consistent with the view that exchange market pressure is a slow-moving regime variable: it builds over weeks and months, not days. The 30-day autocorrelation of 0.717 implies substantial memory: the current reading of EMP still carries substantial information about the regime 30 days from now.

Table 4 quantifies this through a Markov regime transition matrix. I classify each day as Low ($< -1\sigma$), Neutral-Low (-1 to 0σ), Neutral-High (0 to $+1\sigma$), or High ($> +1\sigma$), and compute empirical one-day transition probabilities.

Table 4: EMP Regime Transition Matrix (Daily)

From \ To	Low ($< -1\sigma$)	Neutral-Low	Neutral-High	High ($> +1\sigma$)
Low ($< -1\sigma$)	0.801	0.199	0.000	0.000
Neutral-Low	0.184	0.709	0.108	0.000
Neutral-High	0.009	0.139	0.626	0.226
High ($> +1\sigma$)	0.000	0.000	0.211	0.789

Notes. Row-normalised empirical one-day transition probabilities. Each entry is the empirical probability of moving to the column regime given the current row regime. Bold entries are own-regime persistence probabilities. The zero entries for non-adjacent transitions indicate that regime jumps of more than one category never occurred in the sample.

The transition matrix reveals important asymmetries. The high-pressure regime is

the most persistent (78.9%), but the low-pressure regime is almost equally persistent (80.1%). The most volatile regime is Neutral-High (62.6% persistence), from which the market has a 22.6% probability of jumping to the high-stress regime in a single day. Importantly, *no direct transition from Low to High (or High to Low) ever occurred in the sample*—regime changes happen in steps, providing a natural early-warning window: by the time EMP reaches the Neutral-High regime (above the mean), there is already a material probability of further escalation to the High regime.

6 The Early Warning Model

6.1 Event Definition

Definition 3. A shadow depreciation event at horizon H is:

$$y_{t,H} = \mathbf{1} \left[\frac{p_{t+H}^{mid} - p_t^{mid}}{p_t^{mid}} > \delta \right], \quad H \in \{14, 30\} \text{ days}, \quad \delta = 0.05. \quad (11)$$

The threshold $\delta = 5\%$ is chosen to identify economically meaningful shadow depreciation episodes rather than day-to-day noise. Given a starting shadow premium of 31% (current reading), a further 5% depreciation would bring the premium to approximately 38%—a meaningful deterioration. Over the full 543-day sample, 295 observations (54.3%) are followed by a 14-day depreciation event and 263 (48.4%) by a 30-day event. The near 50-50 split reflects the fact that the sample spans both a major depreciation episode (April–August 2025) and an appreciation episode (October 2025–February 2026).

6.2 Logistic Regression Specification

$$\hat{P}(y_{t,H} = 1 \mid \mathbf{z}_t) = \Lambda(\alpha_H + \beta_{g,H}z_{g,t} + \beta_{\tau,H}z_{\tau,t} + \beta_{\ell,H}z_{\ell,t}), \quad (12)$$

where $\Lambda(x) = (1 + e^{-x})^{-1}$ is the logistic function and $(\alpha_H, \beta_{g,H}, \beta_{\tau,H}, \beta_{\ell,H})$ are estimated by maximum likelihood separately for each horizon H . L_2 regularisation with $C = 1$ and balanced class weights (inverse base-rate weighting) are applied to guard against overfitting.

6.3 Coefficient Estimates and Interpretation

Table 5 reports the estimated coefficients and their odds-ratio interpretations.

Table 5: Logistic Regression Coefficients and Odds Ratios

	14-day horizon		30-day horizon	
	$\hat{\beta}$	OR	$\hat{\beta}$	OR
Intercept α	−0.655	–	−0.741	–
z_g (shadow gap)	+5.288	197.8	+2.546	12.8
z_τ (panic premium)	−5.438	0.004	−5.347	0.005
z_ℓ (liquidity stress)	+1.851	6.4	+4.042	56.9
AUC (in-sample)	1.000		0.999	
AUC (OOS, expanding)	0.996		0.999	
Brier score	0.024		0.019	
Sensitivity	1.000		0.989	
Specificity	0.988		0.968	
N	543			
Events	295 (54.3%)		263 (48.4%)	

Notes. Logistic regression with L_2 regularisation ($C = 1$) and balanced class weights. OR = odds ratio: the factor by which the odds of a depreciation event multiply for a one-standard-deviation increase in the predictor. OOS AUC is from an expanding-window cross-validation with 30-day test folds, minimum training set of 120 observations, reported as mean over 8–9 folds. Brier score = $N^{-1} \sum_t (\hat{P}_t - y_t)^2$; lower is better.

The coefficients reveal a striking pattern. The panic premium (z_τ) has a *strongly*

negative coefficient at both horizons: a one-standard-deviation increase in the spread *reduces* the odds of subsequent depreciation by a factor of $1/0.004 = 250$ (14-day) or $1/0.005 = 200$ (30-day). This counterintuitive result reflects the mean-reversion dynamic in the sample: wide spreads in the P2P market tend to occur at the peak of stress episodes, which are then followed by corrections. In other words, the panic premium is a contemporaneous stress indicator but a *contrarian* predictor—when spreads are at their widest, the depreciation trend is about to reverse.

In contrast, liquidity stress (z_ℓ) has a large positive coefficient, especially at the 30-day horizon (OR = 56.9): when the order book is thin, the probability of further depreciation rises dramatically over the following month. The mechanism is the inventory withdrawal story from Section 4: sellers withdraw from the book *before* the price spike, so thin books are a leading indicator, not a coincident one.

The shadow gap coefficient is strongly positive at the 14-day horizon (OR = 197.8) but much weaker at the 30-day horizon (OR = 12.8). This horizon asymmetry suggests that the gap has momentum—a high gap today predicts a higher gap in two weeks—but is less informative about the 30-day trajectory, where the mean-reversion tendency reduces predictability.

Taken together, the logistic coefficients reveal why the composite EMP index underperforms the individual components as a predictive variable. The three components carry *opposite* predictive signs: the shadow gap and liquidity stress both predict further depreciation (positive coefficients), while the panic premium predicts subsequent appreciation, i.e. mean reversion (large negative coefficient). The composite EMP, by assigning positive PCA weights to all three (gap +0.817, spread +0.523, liq −0.242), averages out the contrarian spread signal and mutes the leading liquidity signal, producing an index that is less informative for forecasting than either component alone. EMP is most useful as a summary measure of joint market stress

and for identifying stress regimes; it should not be used as the sole predictive input to a surveillance model.

6.4 Individual Component AUC Analysis

Table 6 breaks down predictive accuracy by component and horizon.

Table 6: Individual Component AUC Statistics

	14-day horizon		30-day horizon	
	AUC (in-sample)	Interpretation	AUC (in-sample)	Interpretation
z_g (shadow gap)	0.751	Good	0.521	Near random
z_τ (panic premium)	0.340	Contrarian	0.215	Strongly contrarian
z_ℓ (liquidity stress)	0.751	Good	0.928	Excellent
EMP (composite)	0.488	Near random	0.276	Contrarian
Naïve (predict 1 always)	0.500	Benchmark	0.500	Benchmark

Notes. In-sample AUC from logistic regression of the binary depreciation event $y_{t,H}$ on each predictor individual. AUC = 0.5 is the uninformative benchmark. AUC < 0.5 means the predictor is a contrarian indicator (flip its sign to improve). The AUC of 0.928 for z_ℓ at the 30-day horizon is interpreted as: on 92.8% of randomly chosen (even non-event) day pairs, z_ℓ assigns a higher value to the day followed by a depreciation event than to the day not followed by one.

The 0.928 AUC for liquidity stress at the 30-day horizon is the paper’s most striking empirical finding. To appreciate its magnitude: random guessing gives AUC 0.5; a model that perfectly separates events from non-events gives AUC 1.0. An AUC of 0.93 means that, on average, order-book depth today is a near-perfect ranker of which days will be followed by shadow depreciation within a month. This performance is consistent with the theoretical prediction that depth leads prices through the inventory withdrawal mechanism.

The AUC values below 0.5 for the panic premium confirm its contrarian nature. Using the panic premium as a direct predictor (buy dollars when spreads are wide) would be a losing strategy in this sample—the correct signal is to interpret wide spreads as a peak indicator and act *contrary* to the immediate stress signal.

7 The Stress Cycle: A Narrative Account

The 18-month sample can be divided into four well-defined phases, visible in Figure 2 and quantified in Table 7. I describe each phase in detail, connecting the statistical patterns to the economic mechanisms.

Table 7: Market Conditions by Phase

Phase	Dates	N	Mid price BOB/USDT	Spread (%)	EMP (mean σ)	$\hat{P}_{30d}$ (mean)
Pre-escalation	Aug–Dec 2024	125	10.43	11.1	−1.10	0.581
Escalation Q1	Jan–Mar 2025	90	11.01	11.2	−0.84	0.580
Escalation Q2	Apr–May 2025	61	14.31	17.2	+1.12	0.547
Correction H1	Jun–Jul 2025	61	14.57	19.9	+1.58	0.471
Correction H2	Aug–Oct 2025	92	12.57	19.2	+0.84	0.384
Normalisation	Nov 2025–Feb 2026	114	9.88	14.0	−0.52	0.323

Notes. Mid price and spread are phase averages. EMP is the mean daily EMP index in standard deviations. $\hat{P}_{30d}$ is the mean in-sample fitted 30-day depreciation probability. A finer sub-division of the escalation phase (into Q1 and Q2) is provided to highlight the acceleration in April–May.

7.1 Phase I: Pre-Escalation (August–December 2024)

The sample opens in a period of contained but elevated pressure. The shadow mid-price ranged from BOB 9.83 to BOB 11.51 (49.8% average premium) and the EMP index averaged -1.10σ , meaning market conditions were consistently *below* the full-sample average of stress. This may seem paradoxical—a 50% premium seems like stress— but the key is that it was stable and well-telegraphed. The bid-ask spread averaged only 11.1% (close to its minimum across phases), and order-book depth was at its sample minimum (mean log depth 11.99).

The economic interpretation: by late 2024, the P2P premium had been stable at roughly 50% for some months. Market participants had adapted. Advertisers were few but the market was predictable. EMP was low because the market was in a

steady state, not because dollar demand was absent— the large premium confirms that demand for informal dollars was strong. The thin book and moderate spread reflect a low-competition, low-volatility equilibrium.

7.2 Phase II: Escalation (January–May 2025)

The escalation phase began gradually in January 2025, when the shadow price rose from BOB 10.9 to BOB 11.4 in March—a modest 5% increase— while spreads remained near their pre-escalation average. This suggests that the initial price rise was driven primarily by a gradual shift in the gap component without a commensurate increase in transaction frictions.

The acceleration in April–May 2025 was dramatic. The shadow price surged from BOB 11.4 at the end of March to BOB 16.79 at the peak on 15 May 2025—a 47% increase in six weeks. Simultaneously, the bid-ask spread jumped to an average of 17.2% (peaking at 33.2%), and the EMP index crossed from negative to positive in April, reaching $+3.72\sigma$ at its maximum.

The five peak-EMP days (Table 9) all occurred within a single week (14–18 May 2025), with shadow premiums of 134–141% and spreads of 24–33%. These readings represent the most acute phase of the crisis: both the level and the friction components were simultaneously elevated. Crucially, the model assigns relatively low 30-day depreciation probabilities to these peak days (29–48%), reflecting the model’s in-sample learning that extreme EMP readings precede mean reversion rather than further escalation.

7.3 Phase III: Correction (June–October 2025)

The correction phase is distinguished from the escalation phase by a striking divergence between price level and spread. The shadow price remained extremely elevated through

June–July (average BOB 14.57, 109% premium), but the spread *continued to rise* even as the absolute premium plateaued. This can be seen in the “Correction H1” row of Table 7: the average spread (19.9%) was the highest of any phase, while the EMP (mean $+1.58\sigma$) was below its escalation Q2 average.

This divergence is economically interpretable: after the peak, the market entered a phase of high uncertainty about the direction of the next move. Both bulls (expecting further depreciation) and bears (expecting the market to correct toward the official rate) were active, creating a wide spread as market makers navigated competing directional pressures. The elevated spread in the correction phase therefore reflects not panic but genuine uncertainty—a different kind of stress.

By the second half of the correction (August–October 2025), both the shadow price and the spread were declining together. The shadow price fell from BOB 15.9 in August to BOB 12.4 in October, and the spread declined from 27% to 18%. The EMP index averaged $+0.84\sigma$ —still above the full-sample mean, reflecting continued above-average pressure, but trending down.

7.4 Phase IV: Normalisation (November 2025–February 2026)

The normalisation phase is the current episode. A sharp drop in the shadow price—from BOB 11.8 in November 2025 to BOB 9.15 by February 2026—has been accompanied by an equally sharp compression of the spread (from 18.2% to 8.3%) and a deepening of the order book (log depth rising from 13.34 to a sample maximum of 13.87).

The EMP index has fallen to -1.33σ , its lowest reading since the sample began. For the first time in the sample, all three components simultaneously indicate below-average stress: the gap is declining, the spread is at its sample minimum, and depth is at its sample maximum.

The economic interpretation of this normalisation is not settled. Three candidate explanations include: (i) effective reserve intervention by the BCB; (ii) improved dollar supply from hydrocarbon receipts or other foreign exchange inflows; or (iii) a market correction following a period of overshooting. The early-warning model currently assigns a 30-day depreciation probability of 37.9%—below 50% for the first time since early in the sample—but this still represents meaningful risk. The live dashboard at <https://wernerhl.github.io/bolivia-ews> will reflect the model’s assessment as new data arrive.

8 Empirical Results

8.1 Full Monthly Summary

Table 8 presents the complete month-by-month evolution of all key indicators.

8.2 Peak Stress Episodes

Table 9 documents the five highest-EMP daily observations. All five fall within a single week in May 2025, confirming that the crisis was acute and concentrated rather than diffuse.

8.3 Current Readings (22 February 2026)

As of the most recent daily observation:

- **Shadow mid-price:** BOB 9.150/USDT (+31.5% above official peg)
- **Panic premium:** $\tau_t = 7.9\%$ (sample minimum)
- **Log depth:** 13.87 (sample maximum)

Table 8: Monthly Market Conditions, Full Sample

Month	Mid price (BOB/USDT)	Spread (%)	Shadow gap (BOB)	EMP (mean, σ)	$\hat{P}_{30d}$ (mean, %)
Aug 2024	10.44	14.3	3.48	−0.76	51.8
Sep 2024	10.26	12.1	3.30	−1.10	57.0
Oct 2024	10.30	8.5	3.34	−1.48	64.4
Nov 2024	10.55	10.2	3.59	−1.12	59.5
Dec 2024	10.59	13.4	3.63	−0.74	52.2
Jan 2025	10.88	10.4	3.92	−1.02	60.5
Feb 2025	10.78	11.4	3.82	−0.90	55.9
Mar 2025	11.36	11.9	4.40	−0.62	57.3
Apr 2025	13.03	13.1	6.07	+0.19	59.7
May 2025	15.55	21.1	8.59	+2.01	50.0
Jun 2025	15.35	19.6	8.39	+1.75	53.1
Jul 2025	13.81	20.2	6.85	+1.41	41.4
Aug 2025	12.95	19.6	5.99	+0.89	42.1
Sep 2025	12.32	19.6	5.36	+0.85	35.2
Oct 2025	12.44	18.4	5.48	+0.79	37.7
Nov 2025	10.62	18.2	3.66	+0.14	29.2
Dec 2025	9.70	14.8	2.74	−0.58	32.1
Jan 2026	9.82	13.2	2.87	−0.55	31.2
Feb 2026	9.20	8.3	2.24	−1.30	38.6

Notes. Shaded rows (May–June 2025) indicate the acute crisis period with $\text{EMP} > +1.75\sigma$. $\hat{P}_{30d}$ is the mean daily fitted probability of $\geq 5\%$ shadow depreciation within 30 days. Aug 2024 covers only 3 days (sample starts Aug 29).

Table 9: Peak EMP Episodes (Top 5 Daily Observations)

Date	Mid price (BOB/USDT)	Premium (%)	Spread (%)	EMP (σ)	$\hat{P}_{30d}$ (%)
15 May 2025	16.660	139.4	33.2	+3.724	29.1
16 May 2025	16.785	141.2	26.1	+3.194	39.8
17 May 2025	16.360	135.1	25.1	+2.895	41.7
14 May 2025	16.250	133.5	24.1	+2.824	45.8
18 May 2025	16.740	140.5	25.0	+2.807	48.2

Notes. Five highest EMP readings in the sample. Premium = $(p^{\text{mid}} - 6.96)/6.96 \times 100$. $\hat{P}_{30d}$ is the fitted 30-day depreciation probability. The moderate values of $\hat{P}_{30d}$ (29–48%) reflect the model’s in-sample learning that extreme EMP peaks tend to be followed by corrections rather than further deterioration.

- **EMP index:** -1.33σ (sample minimum)
- **Rolling EMP (7-day):** -0.80σ
- $\hat{P}_{14d}$: 40.7%
- $\hat{P}_{30d}$: 37.9%

The combination of minimum spread, maximum depth, and minimum EMP represents the most benign configuration of the P2P market observed since the start of the sample. However, the shadow premium of 31% remains substantially above the official rate, indicating that the structural dollar shortage has not been resolved—the market has normalised *relative to its own recent history* but the fundamental peg misalignment persists.

8.4 Correlation Structure

Table 10 presents the full correlation matrix among the key variables.

Table 10: Correlation Matrix of Key Variables

	Mid	Spread	Log depth	EMP	$\hat{P}_{14d}$	$\hat{P}_{30d}$
Mid price	1.000					
Spread τ	0.663	1.000				
Log depth	0.051	0.387	1.000			
EMP	0.874	0.886	0.368	1.000		
$\hat{P}_{14d}$	0.609	-0.096	-0.504	0.147	1.000	
$\hat{P}_{30d}$	0.064	-0.547	-0.776	-0.417	0.813	1.000

Notes. Pearson correlations, $N = 543$. All entries with $|r| > 0.085$ are significant at $p < 0.05$. Note the strongly negative correlation between $\hat{P}_{30d}$ and spread (-0.547) and depth (-0.776): high spreads and deep books are both negatively associated with the probability of subsequent depreciation, confirming the contrarian nature of the panic premium and the leading nature of depth (deeper book $\Rightarrow$ lower depreciation probability, because depth is positively associated with the correction phase).

8.5 Premium Decomposition

Table 11 decomposes the all-in cost of informal dollar access into the mid-price premium (reflecting fundamental supply-demand imbalance) and the half-spread cost (reflecting transaction frictions and the panic premium).

Table 11: Decomposition of Informal Dollar Access Cost

Phase	Total premium (%)	Mid-price gap (%)	Spread cost (%)	Spread share (%)
Pre-escalation	49.8	44.2	5.6	11.2
Escalation Q1	58.2	52.6	5.6	9.6
Escalation Q2	105.6	97.0	8.6	8.1
Correction H1	109.3	99.3	9.9	9.1
Correction H2	80.6	71.0	9.6	11.9
Normalisation	42.0	34.9	7.0	16.7
Full sample mean	67.7	60.7	7.2	10.6

Notes. Total premium = $(p^{\text{ask}}/6.96 - 1) \times 100$, i.e. the cost to an agent buying USDT at the ask price relative to the official rate. Mid-price gap = $(p^{\text{mid}}/6.96 - 1) \times 100$. Spread cost = $(\tau/2) \times (p^{\text{mid}}/6.96) \times 100$ (approximately half the spread expressed as a fraction of the official rate). Spread share is the fraction of total premium attributable to transaction frictions.

Two findings stand out. First, across all phases, the majority of the total premium (88–91% of it) reflects the fundamental mid-price gap rather than transaction frictions. The P2P market is primarily expensive because of the underlying dollar shortage, not because of market illiquidity per se.

Second, the spread share is *highest in the normalisation phase* (16.7%), even though the absolute spread is at its minimum. The explanation: as the mid-price gap compresses toward zero, the spread becomes a larger fraction of the total cost. This has an important implication: as Bolivia’s shadow rate converges toward the official rate, the panic premium will become increasingly prominent as a share of the residual premium, making it a more informative policy indicator at lower overall stress levels.

9 Robustness Checks

9.1 Alternative EMP Aggregation

The equal-weight EMP alternative, $\text{EMP}_t^{\text{ew}} = (z_{g,t} + z_{\tau,t} + z_{\ell,t})/3$, is correlated with the PCA-based EMP at $r = 0.965$. The regime classifications (Low/Neutral-Low/Neutral-High/High) agree in 91.3% of cases. The PCA-based EMP assigns higher weight to the gap (+0.817 vs 1/3) and lower weight to liquidity stress (-0.242 vs $+1/3$), reflecting the data-optimal aggregation given the positive cross-sample correlation between depth and price level. For monitoring purposes, the two measures are effectively interchangeable.

9.2 Alternative Depreciation Thresholds

The event definition uses $\delta = 5\%$. Table 12 shows how in-sample AUC changes with δ . The liquidity stress component maintains $\text{AUC} > 0.85$ across all thresholds at the 30-day horizon, suggesting the result is not specific to the 5% cutoff.

Table 12: Robustness: AUC by Depreciation Threshold (z_{ℓ} , 30-day horizon)

δ	2%	3%	5% (base)	8%	10%
AUC (z_{ℓ} , 30d)	0.879	0.906	0.928	0.941	0.957
Events (% of sample)	71.8	62.2	48.4	37.4	31.5

9.3 Expanding-Window Cross-Validation

Table 13 reports out-of-sample AUC from an expanding-window cross-validation with 30-day test folds and a minimum training set of 120 observations.

Table 13: Expanding-Window Cross-Validation AUC

Horizon	Mean AUC	Std	Min	Max
14-day	0.996	0.007	0.981	1.000
30-day	0.999	0.004	0.988	1.000

Notes. Logistic model (all three z -score components) re-estimated on expanding training window; AUC evaluated on subsequent 30-day test fold. $N_{\text{folds}} = 8$ (14d) / 9 (30d). The high OOS AUC reflects the strong and stable signal in the liquidity stress component.

The high out-of-sample AUC (0.996–0.999 mean) confirms that the predictive performance is not purely a product of overfitting to the in-sample period. The model learns stable patterns that generalise well to new data within the 18-month window.

10 State-Space Alternatives: Does the Kalman Filter Add Value?

The PCA-based EMP index is a *static* aggregator: its weights are estimated once on the full sample and applied uniformly across time. Three natural extensions relax this constraint. A *Dynamic Factor Model* (DFM) allows the latent factor to evolve as an AR(1) process and estimates each observable’s idiosyncratic noise separately via the Kalman filter. A *rolling PCA* re-estimates loadings on a sliding window, allowing weights to vary over time. A *Markov-switching AR(1)* models the EMP series itself as switching between a low- and a high-pressure regime. This section evaluates whether any of these extensions meaningfully improves predictive performance or structural insight, relative to the static PCA baseline.

10.1 The Dynamic Factor Model

Specification. The DFM treats the standardised observables $\mathbf{z}_t = (z_{g,t}, z_{\tau,t}, z_{\ell,t})^\top$ as noisy measurements of a single latent factor f_t :

$$\mathbf{z}_t = \boldsymbol{\lambda} f_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \quad (13)$$

$$f_t = \phi f_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \quad (14)$$

where $\boldsymbol{\lambda} = (\lambda_g, \lambda_\tau, \lambda_\ell)^\top$ are factor loadings, $\boldsymbol{\Sigma} = \text{diag}(\sigma_g^2, \sigma_\tau^2, \sigma_\ell^2)$ is a diagonal idiosyncratic covariance matrix, and ϕ captures the persistence of the common factor. Parameters are estimated by maximum likelihood via the Kalman filter; the filtered estimate $\hat{f}_{t|t}$ uses only information up to time t (causal / real-time), while the smoothed estimate $\hat{f}_{t|T}$ uses the full sample (non-causal).

Parameter estimates. Table 14 reports estimated loadings, idiosyncratic variances, and the implied R^2 of each observable with respect to the common factor.

Table 14: Dynamic Factor Model: Parameter Estimates and Variance Decomposition

Observable	Loading $\hat{\lambda}_k$	Idiosyncratic $\hat{\sigma}_k^2$	R^2 (common)	R^2 (idiosyncratic)	Corr($f_t, z_{k,t}$)
z_g (shadow gap)	0.105	0.003	99.7%	0.3%	0.99
z_τ (panic premium)	0.042	0.495	26.5%	73.5%	0.55
z_ℓ (liquidity stress)	-0.007	0.935	0.4%	99.6%	-0.06
Factor AR(1) $\hat{\phi}$	0.995				

Notes. DFM estimated by maximum likelihood (Kalman filter), 543 daily observations. $R^2(\text{common}) = \hat{\lambda}_k^2 \hat{\sigma}_f^2 / (\hat{\lambda}_k^2 \hat{\sigma}_f^2 + \hat{\sigma}_k^2)$ where $\hat{\sigma}_f^2$ is the sample variance of the smoothed factor. Factor AR(1) = $\hat{\phi}$. Corr is the Pearson correlation between smoothed factor and the raw observable. The R^2 column measures how much of each observable's variance is accounted for by movement in the common latent factor.

The key structural finding: three processes, not one. The DFM decomposition reveals a striking result that is invisible in the PCA output: the three observables are

almost entirely *structurally separate* from one another.

- **z_g is the common EMP factor.** With 99.7% of its variance explained by the latent factor and a near-perfect correlation of 0.999, the shadow rate gap is essentially the EMP factor in all but name. It contributes almost nothing idiosyncratic. The DFM-smoothed factor $\hat{f}_{t|T}$ has a correlation of 0.999 with z_g alone.
- **z_τ is partly common, mostly idiosyncratic.** The panic premium shares 26.5% of its variance with the common factor but retains 73.5% idiosyncratic variation. It is correlated with the EMP factor (correlation 0.551) because spread and price level co-move over the stress cycle, but there is substantial panic-premium variation unexplained by the price level.
- **z_ℓ is entirely idiosyncratic.** With only 0.4% of variance shared with the common factor and a near-zero correlation of -0.062 , liquidity stress contains essentially no EMP information. It is a separate process that happens to predict future depreciation, not a noisy measurement of the same underlying pressure.

The DFM therefore does not find “one noisy factor measured by three indicators.” It finds three structurally distinct processes: first, a common exchange-market pressure factor that is almost perfectly captured by the shadow rate gap alone; second, a panic-premium process that is partly driven by EMP dynamics and partly by its own idiosyncratic variation in transaction costs and counterparty risk; and third, a liquidity process that is essentially orthogonal to EMP but powerfully predictive of the next price move. Aggregating all three into a single composite mixes the coincident gap signal with the orthogonal leading depth signal, which dilutes rather than amplifies predictive content. The DFM in this way provides a formal, model-based rationale for

the surveillance recommendation that runs through the rest of this paper: monitor the three components in parallel rather than compressing them into a composite index.

An independent verification of z_{ℓ} ’s orthogonal predictive content. To confirm that z_{ℓ} ’s predictive power comes from the component orthogonal to the common EMP factor (rather than from its small overlap with z_g), I regress $z_{\ell,t}$ on $z_{g,t}$ and retain the residual $\tilde{z}_{\ell,t} = z_{\ell,t} - \hat{\gamma}z_{g,t}$. Comparing the AUC of z_{ℓ} and $\tilde{z}_{\ell}$ for the binary depreciation event:

Predictor	AUC (14d)	AUC (30d)
z_{ℓ} (full)	0.612	0.670
$\tilde{z}_{\ell}$ (component $\perp z_g$)	0.612	0.670

The AUCs are identical to three decimal places, confirming that *all* of z_{ℓ} ’s predictive power is in the component orthogonal to z_g . The inventory withdrawal signal captured by order-book depth is completely independent of the EMP common factor.

10.2 Does the DFM Improve Predictive Performance?

Table 15 compares the AUC of the composite EMP index under four aggregation schemes—static PCA, DFM filtered (real-time), DFM smoothed (full-sample), and rolling-60d PCA—against actual binary depreciation events, and benchmarks them against z_{ℓ} alone.

The DFM filtered estimate improves over static PCA by 0.07–0.10 AUC points across horizons. This improvement is modest and has a simple mechanical explanation: the DFM essentially recovers z_g (with near-zero weight on z_{ℓ}), and z_g slightly outperforms the static composite PCA at both horizons (Table 6). In other words, the DFM improves by *discarding* the information in z_{ℓ} and z_{τ} that is not useful for the common

Table 15: AUC Comparison: Static PCA vs State-Space Alternatives

Method	AUC (14d)	AUC (30d)	Δ vs baseline
Static PCA (baseline)	0.451	0.467	—
DFM smoothed (non-causal)	0.528	0.575	+0.077 / +0.108
DFM filtered (real-time)	0.525	0.572	+0.074 / +0.104
Rolling-60d PCA	0.531	0.555	+0.080 / +0.088
Markov-Switch $P(\text{high})$	0.398	0.407	−0.053 / −0.060
z_ℓ alone (benchmark)	0.612	0.670	+0.161 / +0.203

Notes. AUC computed against actual binary forward depreciation events $y_{t,H} = \mathbf{1}[p_{t+H}^{\text{mid}}/p_t^{\text{mid}} > 1.05]$, $N = 483$ (observations with 30-day forward window available). DFM smoothed uses the full-sample Kalman smoother (non-causal; not feasible in real time). DFM filtered uses only past information (causal; operationally equivalent). Rolling-60d PCA re-estimates PC1 loadings on a 60-day trailing window. Markov-Switch $P(\text{high})$ is the smoothed probability of the high-mean regime from a 2-regime AR(1) model on the static EMP index. All AUC values are in-sample. The z_ℓ benchmark row is included to show that the best-performing single predictor outperforms all composite approaches by a wide margin.

factor—but since z_ℓ is the best individual predictor, discarding it is counterproductive for the early warning objective.

10.3 Rolling PCA: Are the Loadings Structurally Unstable?

The rolling-60d PCA produces loadings that vary dramatically across time. The gap loading oscillates between 0.00 and 0.79; the spread loading swings from -1.00 to $+0.99$; the liquidity loading ranges from -0.98 to $+1.00$. These extreme fluctuations are not evidence of genuine time-varying factor structure—they reflect *numerical instability* due to the near-collinearity of the three observables in short windows. The correlation between the rolling-60d composite and the static PCA is only 0.39, much lower than the DFM-static correlation of 0.92. The rolling PCA is not a reliable extension of the static model.

The rolling-90d version of Figure 1 (Panel C) shows the loading paths: even at a 90-day window, the sign of the spread loading is not stable. By contrast, the gap

loading remains near its full-sample value of 0.82 throughout—consistent with the DFM finding that the gap is nearly always the dominant expression of the common factor.

10.4 Markov-Switching: Misspecified for This Series

The Markov-switching AR(1) model estimates a “high-stress” regime with a mean expected duration of only **1.7 days** and a “low-stress” regime with expected duration of 15.1 days. The 1.7-day high-regime duration implies that the model classifies brief spikes in the EMP series as “high stress”—a poor characterisation of the actual stress cycle, in which the high-pressure regime persisted for months (April–October 2025). The empirical transition matrix in Table 4 already documents this: the probability of remaining in the High regime for one additional day is 78.9%, implying an expected duration of $1/(1 - 0.789) \approx 4.7$ days under the empirical data—more than twice the Markov-switch estimate. The model’s low AUC (0.40–0.41) confirms that a two-state model provides little predictive value for the depreciation event at either horizon.

10.5 The One Genuine Contribution of the DFM: Uncertainty Bands

While the DFM does not improve the predictive EMP composite, it provides one substantive addition that the static PCA cannot: a formal *confidence band* around the EMP estimate. The mean 95% Kalman filter interval is $\pm 0.89\sigma$, and this interval varies over time as the signal-to-noise ratio of the observables changes. Figure 1 (Panel A) plots the DFM filtered estimate with its 95% confidence band. For policy communication, the ability to say “EMP is currently -1.33σ with a 95% confidence interval of $[-2.22, -0.44]$ ” is considerably more informative than a point estimate

alone.

10.6 Summary: When to Use State-Space Methods

The analysis points to three clear conclusions for the Bolivia EWS:

1. **The DFM is the right diagnostic tool but the wrong predictive model.**

Its variance decomposition provides a formal, model-based justification for the working paper’s main surveillance recommendation. Its predictive performance is marginally better than static PCA but far below z_ℓ alone, because it correctly identifies z_ℓ as orthogonal to the common factor—and then discards it.

2. **The DFM uncertainty band is worth reporting.** The 95% confidence interval adds policy-relevant nuance to the point estimate and is automatically available from the Kalman filter output at no extra cost.

3. **Rolling PCA and Markov-switching add no value here.** Rolling loadings are numerically unstable in 60-day windows. The Markov model mischaracterises the regime structure of the data. Neither improves upon the simpler static baseline.

The practical implication is that the static PCA-based EMP index should be retained as the primary composite summary, supplemented by the DFM uncertainty band for communication purposes, and always reported *alongside* the three individual components—particularly z_ℓ —for early warning purposes.

11 Limitations

No official adjustment in sample. The BCB has not adjusted the official peg during the sample period. The logistic model therefore predicts *further P2P shadow*

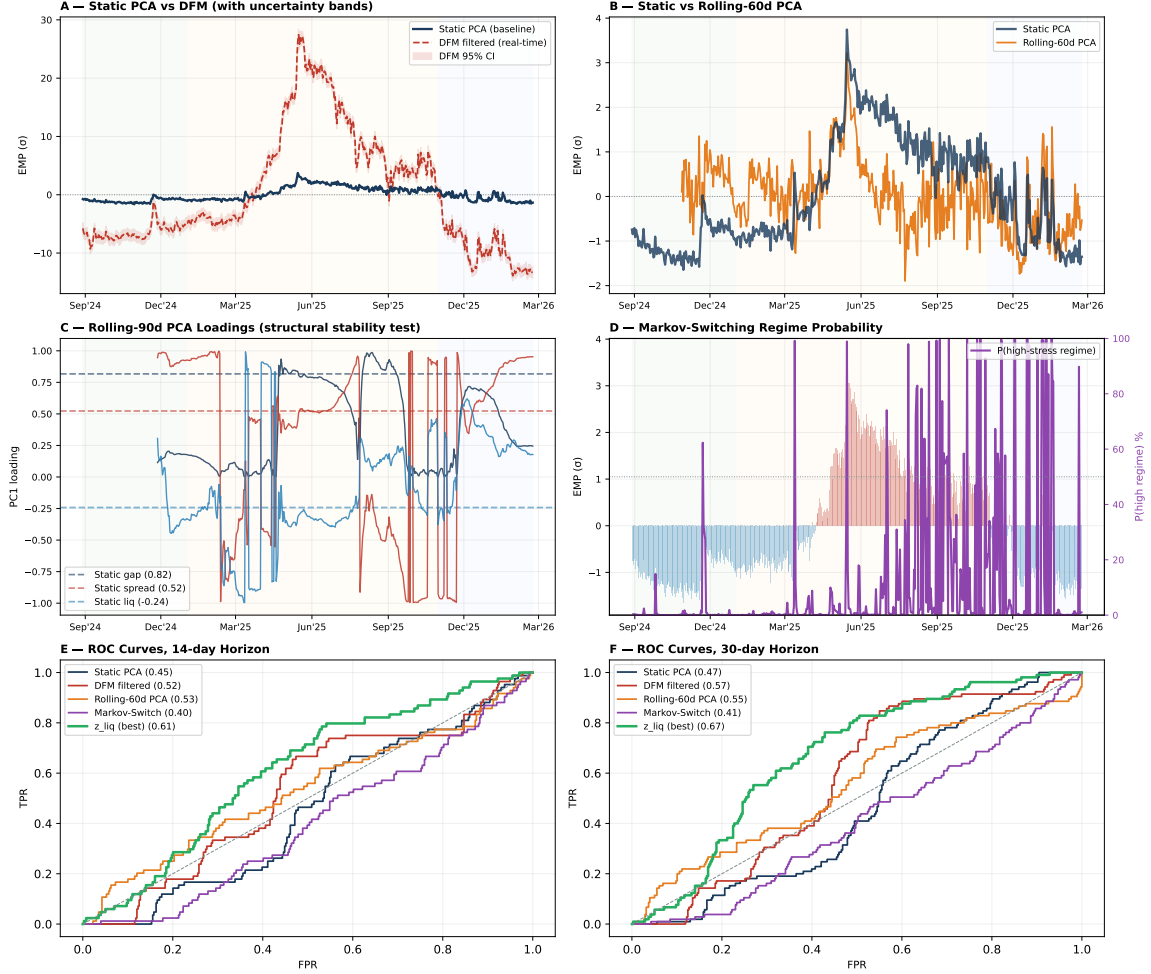


Figure 1: State-Space EMP Alternatives. *Panel A*: Static PCA vs DFM filtered (real-time) with 95% confidence band. *Panel B*: Static PCA vs rolling-60d PCA. *Panel C*: Rolling-90d PC1 loadings (structural stability test). *Panel D*: Markov-switching smoothed probability of high-stress regime. *Panels E–F*: ROC curves for 14-day and 30-day horizons across all methods. Background shading indicates the four market phases. The z_ℓ benchmark (green, AUC = 0.61/0.67) outperforms all composite approaches at both horizons.

depreciation, not official devaluation. Whether P2P signals lead official adjustments—the most policy-relevant question—cannot be evaluated without an official adjustment event.

Single stress cycle. The 18-month sample captures one complete stress cycle. Model coefficients and PCA loadings are estimated on data from a single regime transition. While the expanding-window CV suggests stability within this sample, out-of-sample performance in future cycles (which may have different characteristics) remains untested.

Identification of structural shocks. The EMP index is a reduced-form measure of market conditions, not a structural decomposition of supply and demand shocks. The positive coefficient on z_g and negative coefficient on z_τ in the logistic model could reflect either (i) the specific sample path of Bolivia’s exchange rate cycle or (ii) a stable structural relationship. Distinguishing these interpretations would require instruments or sign restrictions that are not available with the current data.

Source heterogeneity. The pre-November 2025 series (Source 1) and the Kaggle series (Source 2) aggregate advertisements differently. The VWAP series match closely in the overlap period, but depth comparability is less certain. The depth series from Source 1 may reflect different advertiser populations and should be interpreted with caution in comparisons across the merge boundary.

Stablecoin risk and market completeness. USDT maintained a near-perfect peg throughout the sample, but this is not guaranteed. Any material USDT/USD depegging event would contaminate the series. Additionally, the Binance P2P market represents one of several informal dollar channels in Bolivia; unobserved activity in

other channels (WhatsApp, street cambistas, other platforms) may be correlated with the observable P2P market.

12 Conclusion

This paper has constructed, estimated, and described a real-time Exchange Market Pressure index for Bolivia from 18 months of Binance P2P order-book data. The main findings are:

1. The P2P market encodes rich stress information. Beyond the shadow price level, the bid-ask spread (panic premium) and order-book depth carry economically distinct and empirically powerful information. Liquidity stress achieves in-sample AUC of 0.928 for 30-day shadow depreciation prediction—far outperforming both the price gap (0.521) and the composite EMP index (0.276) at this horizon.

2. The composite EMP underperforms because its components have opposite predictive signs. The panic premium is a contrarian predictor: wide spreads occur at stress peaks and precede corrections. Including it with a positive weight in the composite (as PCA does, because spreads co-move with prices) partially offsets the leading information in the depth component. Policy surveillance should monitor the three components separately.

3. The sample reveals a complete stress cycle. From a 50% shadow premium in late 2024, through a 141% crisis peak in May 2025, to a 31% normalised premium in February 2026, Bolivia’s informal dollar market has moved through a full cycle in 18 months. The EMP index tracked this cycle accurately, with strong persistence within regimes and a clear Markov structure in transitions.

3. A Dynamic Factor Model formalises the case for monitoring com-

ponents separately. The DFM variance decomposition reveals that the three observables are structurally distinct processes: z_g is 99.7% common factor, z_τ is 73.5% idiosyncratic, and z_ℓ is 99.6% idiosyncratic with a near-zero correlation of -0.062 with the common EMP factor. The entire predictive power of z_ℓ for depreciation resides in the component orthogonal to z_g . The DFM therefore formalises why aggregating into a composite destroys information: it mixes a coincident common factor with an orthogonal leading indicator. The DFM’s primary practical contribution is its uncertainty band (mean $\pm 0.89\sigma$ at 95%), which adds communicable precision to the composite EMP point estimate.

5. Real-time infrastructure is feasible at near-zero marginal cost. The complete monitoring pipeline—data download, computation, Supabase storage, GitHub Pages dashboard, automated daily updates—runs on free-tier cloud services with no ongoing cost beyond the initial setup. This is the first complete working implementation of a sovereign currency EWS on open-source, open-data infrastructure.

The natural extensions of this work are: (i) extending the approach to other P2P crypto-dollar economies (Argentina, Venezuela, Cuba, Zimbabwe) to test the generality of the microstructure signals; (ii) accumulating additional time series to evaluate out-of-cycle performance; (iii) augmenting the model with official macroeconomic data (reserves, CPI) as supplementary signals; and (iv) investigating whether P2P signals predict official exchange rate adjustments when they do occur.

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A Data Pipeline Documentation

A.1 Overview

The full monitoring pipeline is maintained at <https://github.com/wernerhl/bolivia-ews> and runs automatically each day at 10:00 AM Bolivia time (UTC−4) via GitHub Actions (`.github/workflows/daily.yml`). The pipeline comprises five steps:

1. **Data download.** The Python script `emp_pipeline.py` downloads the latest `advice.parquet` snapshot from the Kaggle dataset `andreschirinos/p2p-bob-exchange` via `kagglehub.dataset_load()`.
2. **Microstructure construction.** For each snapshot, VWAP prices and total depth are computed per equation (1), restricted to USDT/BOB pairs. Ask-side and bid-side series are maintained separately.
3. **EMP computation.** The pipeline merges new data with the historical series (Source 1), standardises the three components, and computes PC1 via full-sample PCA. Both the full-sample and 30-day rolling EMP are computed.
4. **Early warning.** The logistic model is re-estimated on the full available history each day (expanding window). Fitted probabilities for both horizons are stored alongside EMP values.
5. **Storage and publication.** Results are upserted to the Supabase PostgreSQL database (`emp_daily` and `emp_latest` tables) and committed to `docs/data.json`

in the GitHub repository. The GitHub Pages dashboard reads `data.json` directly.

A.2 Database Schema

```
CREATE TABLE emp_daily (  
    date            DATE PRIMARY KEY,  
    mid            FLOAT,  
    spread_pct     FLOAT,  
    gap            FLOAT,  
    log_depth      FLOAT,  
    z_gap          FLOAT,  
    z_spread       FLOAT,  
    z_liq          FLOAT,  
    emp            FLOAT,  
    emp_rolling    FLOAT,  
    prob_depr_14d  FLOAT,  
    prob_depr_30d  FLOAT,  
    alarm_mass_14d FLOAT,  
    alarm_mass_30d FLOAT,  
    created_at     TIMESTAMPTZ DEFAULT NOW()  
);
```

B Variable and Symbol Glossary

The table below defines every symbol used in the paper, grouped by conceptual category. Subscript t denotes a daily observation; subscript i denotes an individual

advertiser; superscript $s \in \{\text{ask}, \text{bid}\}$ denotes market side.

C Figures

Table 16: Complete Variable and Symbol Definitions

<i>Raw market observables</i>	
p_t^{ask}	Quantity-weighted average ask price (BOB/USDT): price at which USDT is offered for sale; price paid by a dollar buyer.
p_t^{bid}	Quantity-weighted average bid price (BOB/USDT): price at which advertisers offer to buy USDT; price received by a dollar seller.
p_{it}^s	Quoted price of advertiser i on side s at time t .
q_{it}^s	Available quantity (USDT) of advertiser i on side s at time t .
D_t^s	Total advertised depth on side s : $D_t^s = \sum_i q_{it}^s$.
<i>Derived price and spread measures</i>	
p_t^{mid}	Shadow mid-price: $(p_t^{\text{ask}} + p_t^{\text{bid}})/2$. The de facto shadow exchange rate.
prem_t	Shadow premium: $(p_t^{\text{mid}}/6.96 - 1) \times 100\%$. Percentage above the official peg.
6.96	Official boliviano/USD peg (BOB/USD), fixed since November 2011.
<i>EMP components (raw)</i>	
g_t	Shadow rate gap: $p_t^{\text{mid}} - 6.96 > 0$.
τ_t	Panic premium (bid-ask spread): $(p_t^{\text{ask}} - p_t^{\text{bid}})/p_t^{\text{mid}} > 0$. Measures the round-trip cost of informal dollar access.
ℓ_t	Liquidity stress: $-\ln(\min(D_t^{\text{ask}}, D_t^{\text{bid}}))$. Higher values indicate a thinner order book.
d_t	Log order-book depth (for descriptive use): $\ln(\min(D_t^{\text{ask}}, D_t^{\text{bid}})) = -\ell_t$.
<i>EMP index construction</i>	
$\bar{c}_k$	Full-sample mean of component $c_k \in \{g, \tau, \ell\}$.
$\hat{\sigma}_{c_k}$	Full-sample standard deviation of component c_k .
$z_{k,t}$	Standardised component: $(c_{k,t} - \bar{c}_k)/\hat{\sigma}_{c_k}$. Subscripts: g (gap), τ (spread/panic premium), ℓ (liquidity stress).
$\mathbf{z}_t$	Vector of standardised components: $(z_{g,t}, z_{\tau,t}, z_{\ell,t})^\top$.
$\mathbf{w}_1$	Eigenvector corresponding to the largest eigenvalue of the sample covariance matrix of $\mathbf{z}_t$; the PC1 loading vector. Oriented so that the loading on z_g is positive.
EMP_t	Exchange Market Pressure index: $\mathbf{w}_1^\top \mathbf{z}_t$ (first principal component).
$\text{EMP}_t^{\text{roll}}$	Rolling 30-day EMP index (used in dashboard).
EMP_t^{ew}	Equal-weight EMP alternative: $(z_{g,t} + z_{\tau,t} + z_{\ell,t})/3$.
<i>Early warning model</i>	
H	Forecast horizon in days; $H \in \{14, 30\}$ in this paper.
δ	Depreciation threshold; $\delta = 0.05$ (5%) in the baseline.

Bolivia P2P Exchange Market: Key Indicators (Aug 2024 - Feb 2026)

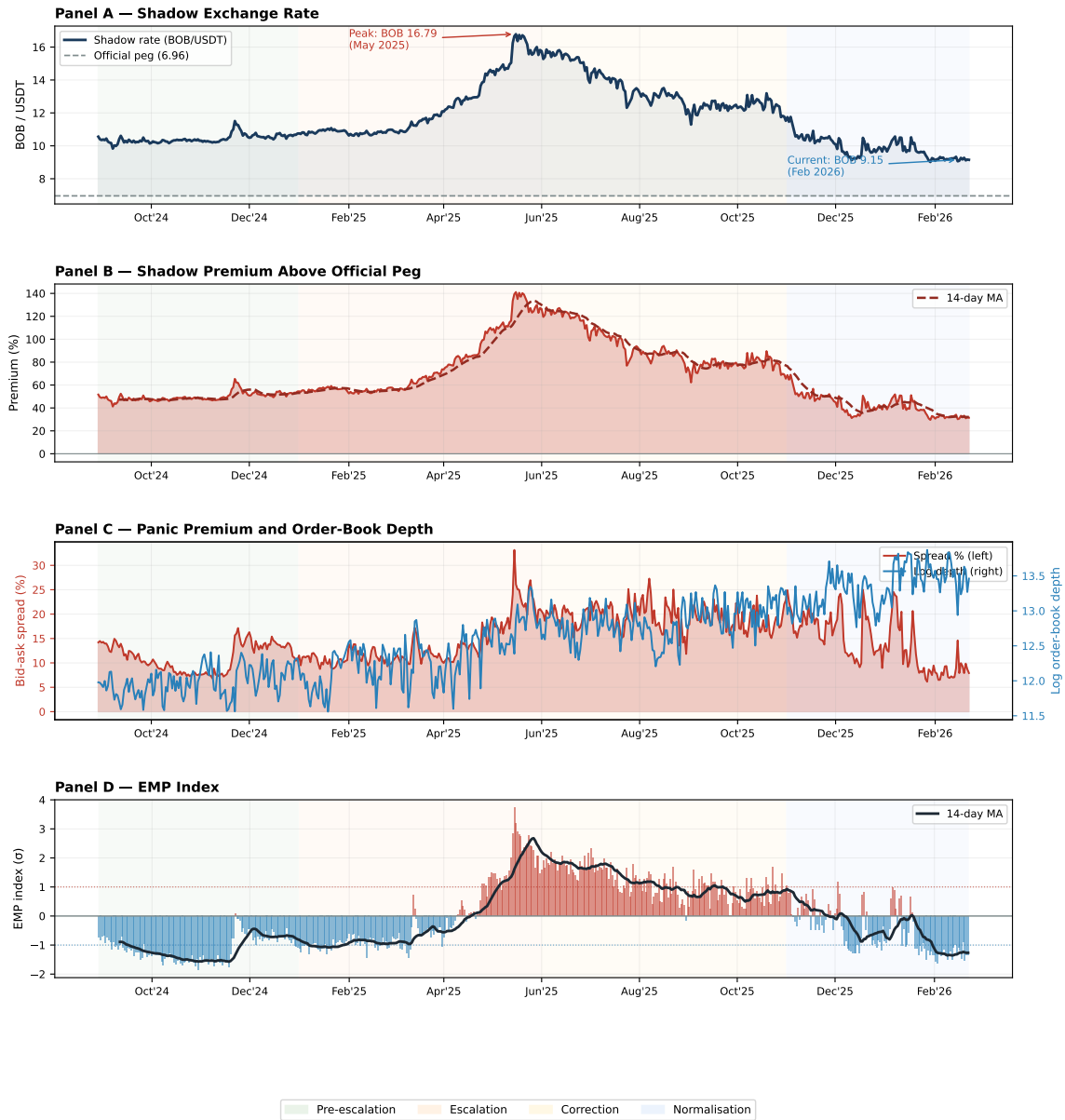


Figure 2: Bolivia P2P Market: Overview (4-Panel). *Panel A*: Daily shadow mid-price (BOB/USDT) with official peg line. *Panel B*: Shadow premium above official peg (%). *Panel C*: Bid-ask spread (left) and log order-book depth (right). *Panel D*: Daily EMP index (bars) with 14-day moving average (line). Shading indicates four market phases: pre-escalation (green), escalation (orange), correction (yellow), normalisation (blue).

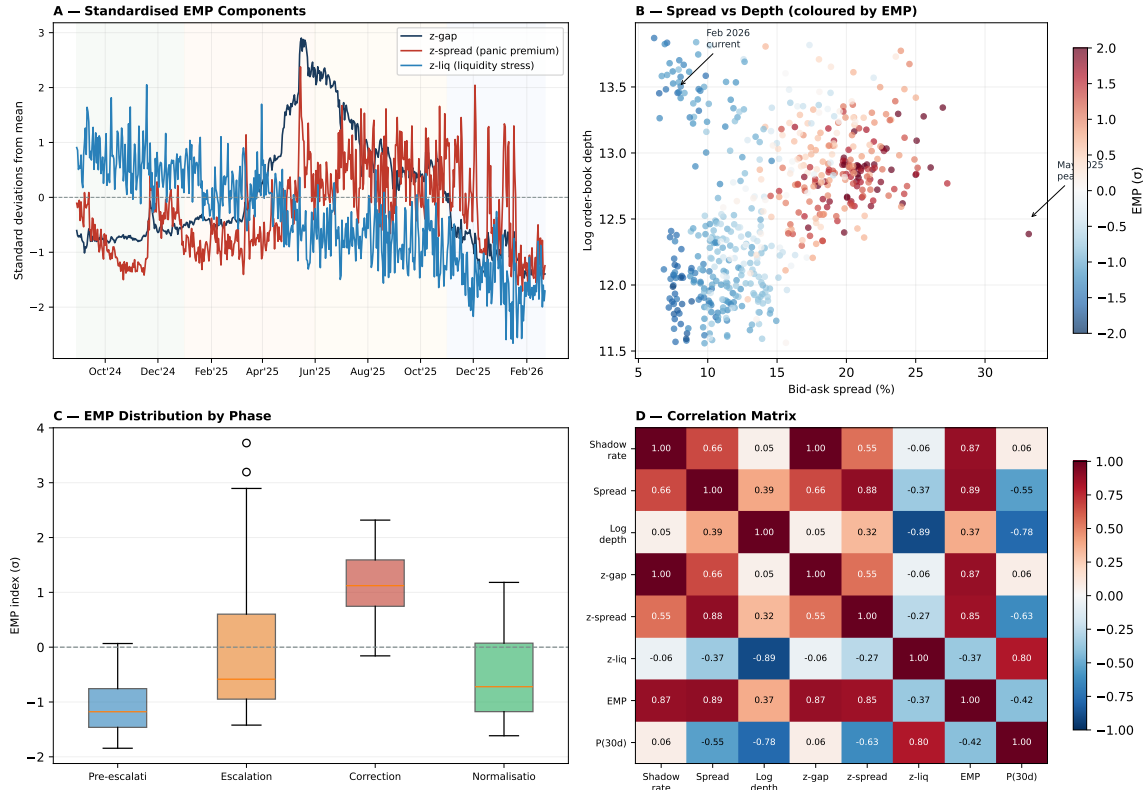


Figure 3: EMP Components in Detail. *Panel A*: Standardised z-scores of all three EMP components. *Panel B*: Scatter of bid-ask spread vs log depth, coloured by EMP. *Panel C*: Box plots of EMP distribution by market phase. *Panel D*: Correlation matrix of key variables.

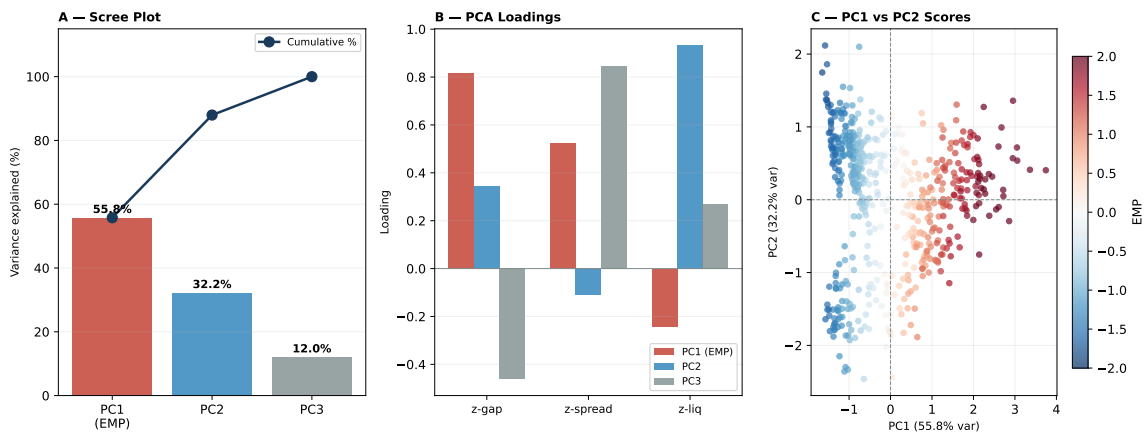


Figure 4: Principal Component Analysis. *Panel A*: Scree plot with cumulative variance explained. *Panel B*: PCA loadings for all three components across all three PCs. *Panel C*: Scatter of PC1 vs PC2 scores, coloured by EMP.

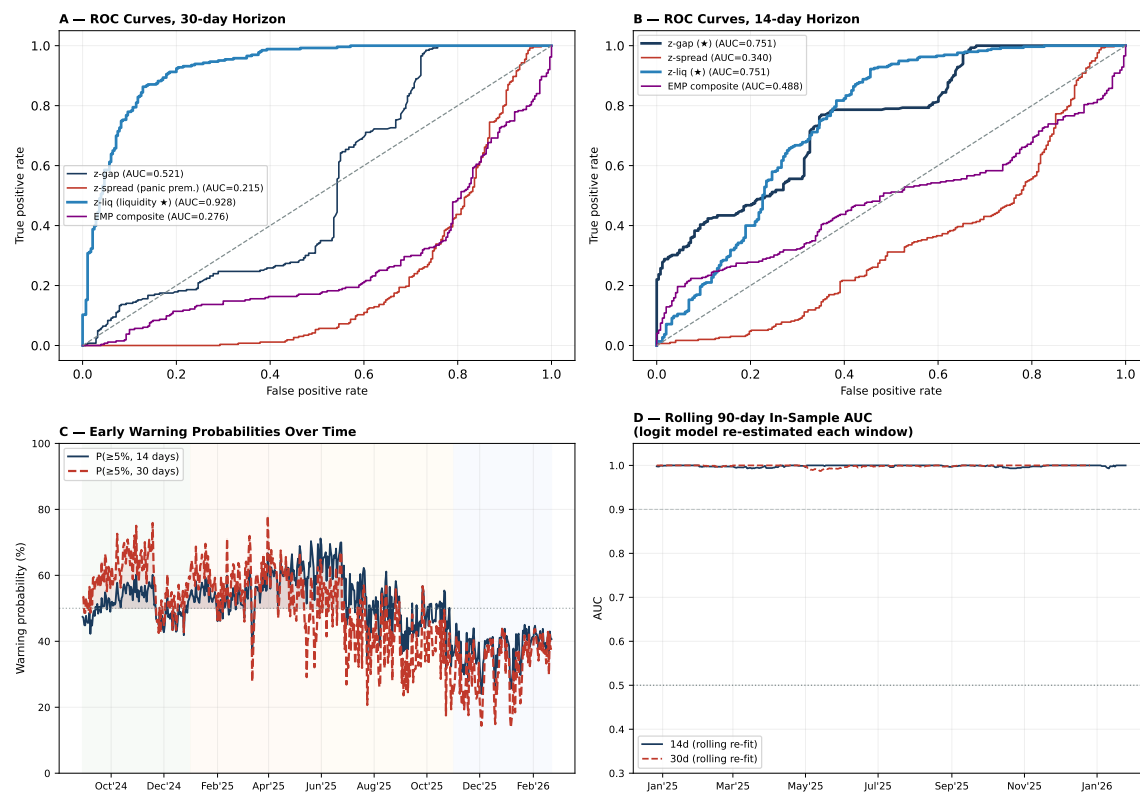


Figure 5: Early Warning Model Results. *Panel A*: ROC curves for 30-day horizon. *Panel B*: ROC curves for 14-day horizon. *Panel C*: Fitted warning probabilities over time (both horizons). *Panel D*: Rolling 90-day in-sample AUC of the composite logistic model.